# Project Design Phase-II
## Technology Stack (Architecture & Stack)

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|---|---|---|---|
| **Date** | Not specified | **Team ID** | Not provided |
| **Project Name** | Global Fuel Prices & Taxation (2020–2026) | **Maximum Marks** | 2 Marks |

### Define Your Technology Stack:

This project uses Tableau Desktop to analyze fuel-price and taxation data from an Excel workbook. The workbook contains country, region, income level, date, fuel prices, tax percentage, subsidy level and Brent crude price fields. The selected tools support data checking, aggregation, comparison and dashboard reporting.

| S.N | Component | Description | Technology | Justification / Purpose |
|---|---|---|---|---|
| 1. | Programming / Query Language | Language and calculations used to aggregate, compare and analyze fuel-price and taxation data. | Tableau calculated fields, built-in aggregations (AVG), sorting and date fields. | Supports average petrol, diesel and LPG prices, average tax percentage, regional comparisons and trends over time. |
| 2. | Data Source / Storage | The structured file used to store the project data before analysis and visualization. | Excel workbook (.xlsx) connected directly to Tableau Desktop. | Contains Country, Region, Income Level, Date, Petrol Usd Liter, Diesel Usd Liter, Lpg Usd Liter, Tax Percentage, Subsidy Level and Brent Crude Usd fields. |

## Technology Stack (continued)

The following components describe how data is loaded, prepared, visualized and presented in the project.

| S.N | Component | Description | Technology | Justification / Purpose |
|---|---|---|---|---|
| 3. | Data Ingestion | Process used to bring the source data into the analysis tool. | Tableau Desktop data connection to the Excel workbook. | Loads the workbook fields into Tableau so they can be used across worksheets and dashboards. |
| 4. | Data Preparation & Business Logic | Review and prepare fields for analysis, including data types, geographic roles, dates and aggregation choices. | Tableau Data Source page, field data types, geographic role settings and AVG aggregations. | Helps ensure prices are compared as averages rather than misleading totals, and supports the 2020–2026 date trend. |
| 5. | Data Visualization / Analytics | Visual methods used to explore country-level values, regional differences, changes over time and relationships between measures. | Tableau maps, horizontal bar charts, line chart and scatter plot. | Displays the global petrol-price map, regional petrol/diesel/LPG comparisons, fuel-price trends, tax comparison, income-level comparison and taxation versus petrol price. |
| 6. | Dashboard / User Interface | The interface through which users explore the charts and compare the project findings. | Tableau dashboard and Tableau Story in the workbook. | Brings the worksheets together in a global overview. Region and Income Level filters allow users to focus on selected groups and interact with the views. |